

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: ARTIFICIAL INTELLIGENCE

COURSE CODE: CSA17

ANALYTICAL PROBLEM SOLVING – PROBLEM STATEMENTS

1. Water Jug Problem

You are given 2 jugs, a 4-gallon one and a 3-gallon one. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into the 4-gallon jug?

Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another, and no other measuring devices are available.

2. Mars Rover – Intelligent Agent Analysis

Find out how the Mars Rover analyses and explores samples on Mars by answering the following questions:

- What are the percepts for this agent?
- Characterize the operating environment.
- What are the actions the agent can take?
- How can one evaluate the performance of the agent?
- What sort of agent architecture do you think is most suitable for this agent?

3. The 8-Queens Problem

Explore the search space using search strategies and formulate the problem components for the 8-Queens problem: place 8 queens on a chessboard such that none of the queens attack any of the others. A configuration of 8 queens on the board is shown below, but this does not represent a solution, as two of the queens share a diagonal.

Column	1	2	3	4	5	6	7	8
Row	1	5	8	4	2	7	3	6

4. OLA Cab Booking – Problem-Solving Agent

A customer wants to travel from one location to another using the OLA Cab booking mobile application. The customer can select any of the cab types – Mini, Micro, Sedan, Shared, Prime, and so on – based on comfort, to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. Logistics Network – Uniform Cost Search (UCS)

A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm. The delivery network is represented as a weighted graph connecting nodes S, A, B, C, D and G, as shown in the assessment figure.